Savannah L. Ferretti

savannah.ferretti@asu.edu | savannahferretti.github.io | savannahferretti | savannahferretti

EDUCATION

Ph.D. in Earth System Science, University of California, Irvine — Jun 2026

M.S. in Earth System Science, University of California, Irvine — Dec 2023

B.S. in Earth & Atmospheric Sciences, Cornell University — Dec 2020

EXPERIENCE

Postdoctoral Researcher — Nov 2026–Present
Kerner Group, Arizona State University — Tempe, AZ

Postdoctoral Researcher — Aug–Oct 2026
Pritchard & Baldwin Groups, University of California, Irvine — Irvine, CA

- Advanced ongoing research on South Asian monsoon rainfall variability, focusing on manuscript development and publication

Graduate Student Researcher — Sep 2021–Jun 2026
Pritchard & Baldwin Groups, University of California, Irvine — Irvine, CA

- Used data-driven and physics-based approaches to identify & explain drivers of South Asian monsoon rainfall variability
- Developed an interpretable machine learning methodology that used neural networks & equation discovery to learn physically meaningful relationships from high-dimensional atmospheric data
- Built open-source Python pipelines on GitHub for cloud data access, large-array processing, uncertainty analysis, model evaluation, & visualization in HPC environments
- Communicated findings through peer-reviewed publications & conference presentations

Consulting Analyst — Mar–Aug 2021
Higher Education Research Services, Huron Consulting Group — New York, NY

- Built Excel-based financial planning & reporting tools for a research nonprofit
- Developed long-term resource forecasts & presented recommendations to executive stakeholders

SKILLS

Python (PyTorch, PySR, Dask, Xarray, Cartopy, Matplotlib) • GitHub • High-Performance Computing • NetCDF/HDF5/Zarr • Bash • Linux • Microsoft Excel (Certified Specialist) • LaTeX • WordPress

SELECTED PUBLICATIONS

S. L. Ferretti, J. Lin, S. Shamekh, J. W. Baldwin, M. S. Pritchard, & T. Beucler. (2026). Data-driven integration kernels for interpretable nonlocal operator learning. *Environ. Data Sci.*

S. L. Ferretti, M. S. Pritchard, F. Ahmed, L. Peng, & J. W. Baldwin. (2025). Explaining South Asian monsoon rainfall seasonality using a metric of plume buoyancy. *Geophys. Res. Lett.*